

Building a Single-Speaker, Emotion-Tagged TTS Dataset

Indian English + Telugu, sourced from YouTube, with Sarvam APIs

1. What I built and how the pipeline works

The goal was a **high-quality TTS dataset of single-speaker segments** — ~30 minutes of Indian English and ~30 minutes of Telugu — with accurate transcriptions and per-segment emotion/style tags, published publicly on HuggingFace. (*"Single-speaker" means each segment contains exactly one speaker; the dataset spans 11 distinct speakers total — 5 English, 6 Telugu — tracked via `speaker_id`, which is more useful for TTS than one voice.*) The grading emphasis is on *data-quality judgment and curation*, not scripting, so I treated the code as a tool in service of curation, and spent the real effort on **source selection** and on a **listen-and-iterate** loop.

The pipeline is a small, modular Python package (`ttlds`) driven by a typer CLI. It processes **one source at a time** so quality can be validated (and API credits protected) before scaling. Stages:

1. **Source curation** (`config/sources.yaml`) — the highest-leverage step. Candidate YouTube sources are chosen by archetype (audiobooks, solo lectures, AIR talks, storytelling, TEDx) and channel reputation, deliberately avoiding compilation channels that overlay music. A discovery aid (`scripts/discover.py`) searches YouTube metadata to surface candidates; every one is still validated downstream and by ear.
2. **Download + normalize** — `yt-dlp` pulls best-quality audio plus provenance (channel, upload date, license, video id). `ffmpeg` produces two derivatives: a **16 kHz mono** copy for ASR and a **24 kHz mono master** for the published audio.
3. **Batch ASR + diarization** — Sarvam `saaras:v3` batch STT with `with_diarization + with_timestamps` gives speaker-labelled, time-stamped chunks. This provides *structure*: who spoke when.
4. **Single-speaker, silence-snapped segmentation** — the dominant speaker's chunks are merged into runs (a different speaker breaks a run, so clips never span a speaker change). Runs are split into **3–25 s** clips (target 5–15 s) with cuts **snapped to silences** via local energy detection, never mid-word.
5. **Per-segment realtime transcription** — each finished clip is re-transcribed with Sarvam `saarika:v2.5`. The batch transcript is coarse; this second pass yields a transcript exactly aligned to the published clip, plus word timing (→ speaking rate) and a language-confidence signal. This **double-pass ASR** is a deliberate quality investment.
6. **Acoustic features** — `parselmouth` (Praat) + `librosa` extract pitch (F0 mean/range/variation), energy/RMS dynamics, harmonics-to-noise ratio (HNR), high/low spectral band ratio (breathiness), voicing fraction, and pauses. Features are **z-scored per speaker**, so "excited" means *elevated for that voice*, not an absolute threshold.
7. **Quality gates** — each clip gets metrics + a status (pass / flag / reject) with explicit reasons: sustained clipping, low SNR, excessive silence, a music/noise bed (gap-energy ratio), low ASR confidence, implausible character rate, and near-duplicate transcripts. Flags (e.g. music bed, low tag confidence) are kept but surfaced for human review; rejects are dropped.

8. **Emotion + style tagging** — a hybrid of acoustics and an LLM. The per-speaker acoustic profile is rendered into plain language and sent, with the transcript, to Sarvam **sarvam-30b** constrained to a closed taxonomy. **Whisper is decided by an acoustic rule** (low voicing + low HNR + low energy), not LLM text-guessing, and the LLM is instructed to ground emotion in the acoustics (flat prosody → neutral even if the words are emotional).
9. **Human-in-the-loop review** (`ttsds review-build`) — a self-contained static HTML app lists every candidate with an audio player, editable transcript, emotion/style dropdowns, and all metrics. The reviewer accepts/rejects/relabels; **a human edit always overrides the automated tag** (`tag_source` flips to human).
10. **Balance + finalize** — accepted clips are selected to hit ~30 min/language while **balancing the emotion histogram** (round-robin across emotion buckets so rare emotions are fully included and neutral is capped). Audio gets **light loudness normalization** (≈ -20 LUFS, peak -1 dBFS) *without* limiting, so the prosodic dynamics that carry emotion are preserved.
11. **Publish + verify** — the dataset is built with HuggingFace `datasets` (two configs, `indian_english + telugu`, each with train/validation), pushed **public**, and reloaded with `load_dataset` to confirm audio decodes and the schema is intact.

Key design decisions (the judgment that matters)

- **Double-pass ASR** — diarization for structure, realtime re-ASR for clip-accurate text.
- **Silence-snapped cuts** — never trust coarse chunk boundaries; cut in pauses.
- **Per-speaker-relative emotion** — z-scores, not absolute prosody thresholds.
- **Acoustically-grounded emotion + acoustic whisper override** — the LLM can't text-guess prosody.
- **Light normalization** — aggressive -23 LUFS limiting would flatten the very dynamics being tagged.
- **Human override is final** — the dataset's labels are only as good as what a person confirms.
- **Incremental, per-source validation** — catch problems on one source before spending credits on twelve.

2. Iterations to improve data quality

The pipeline only looked finished after running it on real audio and *looking at the output*. Three concrete iterations, each found by inspecting iteration-0 results:

Iteration A — the clipping gate was rejecting clean audio. On the first Telugu audiobook, **41 of 43 candidate segments were rejected**, all for "clipping". Inspection showed the audio was actually pristine (SNR 24–30 dB, no music bed, clean transcripts) — but YouTube audio is mastered hot, so peaks *touch* full scale. My gate rejected any peak > 0.99 . Measuring the actual **clipped-sample fraction** showed ≤ 0.03 % of samples at full scale — nowhere near audible clipping (real flat-topping is > 1 %). Fix: gate on *sustained* clipped fraction (> 1 %), not peak. Result: 43/43 kept.

Iteration B — the LLM was parroting my prompt template. With clipping fixed, every clip came back `neutral/narrative`. The cause: 34/43 responses literally copied my JSON example — `"rationale": "<=20 words"`, confidence: `0.0` — instead of classifying. A copyable placeholder template plus low reasoning effort made the model echo the example. Fix: removed the literal template, described the fields instead, and explicitly told the model to pick real values.

Iteration C — the reasoning model was truncating before it answered. sarvam-30b/105b are reasoning models that emit ~1.6–2.3k tokens of reasoning *before* the answer. At max_tokens=1500, **7/8 responses truncated** (finish_reason: length) with empty content, falling back to a default tag. Raising the ceiling to 4000 (billed on actual tokens, so the ceiling is free insurance) gave **8/8 valid, varied, confident tags**. After this, the audiobook produced a rich emotion mix (sad, excited, angry, calm, happy, neutral, fearful) at median confidence 0.85 with zero fallbacks.

Other refinements made along the way: a **gap-energy** music-bed detector (residual energy in inter-phrase pauses) instead of an ambiguous spectral-flatness threshold; **parallelized** the per-segment ASR and LLM calls (the reasoning model is slow sequentially); pinned a **Python 3.12** environment because key audio libraries lack 3.14 wheels.

Three operational issues surfaced near the finish and were worked through: - **Credit exhaustion mid-run.** The Sarvam quota ran out while processing Telugu (4 sources failed with insufficient_quota_error). Because the pipeline is per-source and idempotent, topping up the key and re-running process-all --lang te cleanly recovered exactly the missing sources — no reprocessing of finished ones. - **datasets 4/5 requires torchcodec** to encode the Audio feature, which is painful on Windows. Pinned datasets<4 (3.x encodes via soundfile), verified before publishing. - **Split-config bug in the dataset card.** The first push wrote all clips into a single train split because the card's data_files glob (config/**) matched both train-* and validation-* parquets. Fixed by mapping splits explicitly in the card YAML and re-uploading — train/validation then resolved correctly.

3. What worked and what didn't

Worked well - Sarvam batch diarization gave clean single-speaker structure on solo audiobooks/lectures. - Audiobook narration was the best source type: clean, single voice, wide emotional range. - The double-pass ASR produced clip-accurate transcripts; Telugu transcription quality was strong. - Per-speaker z-scoring made emotion tags coherent within a voice. - The objective gates (clipping fraction, SNR, gap-energy) were a reliable first filter; the HTML review app made human verification fast.

Didn't work / needed care - Naïve thresholds (peak-based clipping) over-rejected clean audio — only visible by inspecting data. - Reasoning-model quirks (template parroting, token truncation) silently degraded tags until inspected. - Compilation/"motivational" channels almost always carry a music bed — excluded at the source step. - TEDx/discourse sources can include applause/audience; the gates flag these for review.

4. Results and quality observations

Published dataset: <https://huggingface.co/datasets/AkCodes23/sarvam-tts-in-te-en> (two configs, each with train/validation).

	Indian English	Telugu
Minutes	30.05	30.25
Clips (train/val)	142 (134/8)	140 (133/7)
Distinct speakers	5	6

Emotion mix	neutral 31, calm 31, sad 30, excited 30, angry 12, fearful 5, happy 3	calm 25, neutral 24, angry 24, excited 24, sad 24, fearful 9, happy 7, surprised 3
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Total: 60.3 minutes. Sources span audiobooks, solo lectures, All India Radio talks, storytelling, a TEDx talk, and discourse — 11 sources, all single-speaker.

Funnel: 457 candidate segments → **445 kept** → balanced down to **282** selected. Only **12 were rejected, all for low SNR** (from one noisier AIR source); clipping and music-bed rejections were zero after the gate fixes. Per-source medians: SNR 19–45 dB, inter-pause gap-energy 0.00–0.03 (no music beds anywhere — even the kathalu/motivation sources I'd flagged as risky came back clean), emotion-tag confidence 0.85. The round-robin balancer capped dominant emotions (Telugu sad 73→24, excited 65→24) and included rare ones fully (surprised 3, happy 7). - **Emotion validity** is the hardest dimension. Grounding the LLM in per-speaker acoustics (not text alone) and overriding whisper acoustically materially improved it, but subtle affect remains imperfect — hence the human review pass and the `emotion_confidence` + `tag_source` fields shipped with every row for transparency. - **Licensing/ethics**: clips are short and transformative, used for research; full per-clip provenance (`source_url`, `source_channel`, `license`) is retained. Permissive / government / educational sources (AIR, NPTEL) were preferred.

5. Evaluation — proving the quality claims

A reviewer shouldn't take "the data is good" on faith. This section provides *evidence*. (I can't subjectively listen to audio, so transcripts and emotion are validated with independent, automated cross-checks; the human-review tool captures true human judgments as the natural next layer.)

5.1 Is it actually single-speaker? — speaker-embedding verification

Diarization is not proof. I embedded all **445 kept clips** with **ECAPA-TDNN** (`speechbrain/spkrec-ecapa-voxceleb`) and compared cosine similarity within vs. between the assigned `speaker_ids`:

Metric	Value
Avg. same-speaker similarity	0.737
Avg. different-speaker similarity	0.213
Separation	0.524 (> 0.3 is strong)
Speakers flagged for contamination	0 / 11

Every speaker's clips are far closer to each other than to any other speaker (see the speaker-similarity heatmap). Objective evidence that each `speaker_id` is one consistent voice — each *segment* is single-speaker, across 11 voices total.

5.2 How accurate are the transcripts? — cross-ASR agreement

True WER needs human references; instead I re-transcribed a deterministic subset with **Whisper** (an unrelated ASR) and measured divergence from the Sarvam transcripts with `jiwer`. Low divergence ⇒ two independent systems agree ⇒ high reliability.

Reference ASR	Whisper small.en	Whisper large-v3
WER (Sarvam vs Whisper)	6.8 %	74.9 %
CER	4.5 %	34.6 %

English: 6.8 % WER / 4.5 % CER means two *independent* ASRs agree closely — strong evidence the English transcripts are reliable. **Telugu:** the high divergence is **Whisper's limitation, not Sarvam's** — Whisper is weak on Telugu (under-resourced in its training), so this number measures the *reference*, not the dataset; cross-ASR is therefore not a valid Telugu transcript-quality proxy, and I won't pretend it is. Two clean supporting signals for Telugu: the realtime ASR identified the correct language (te-IN) on **100 %** of clips (and en-IN on 100 % of English clips — garbled audio wouldn't language-ID correctly), and Sarvam's models are purpose-built for Indic. A definitive Telugu transcript audit needs **human review** — the review tool supports exactly that, and Telugu was chosen so a fluent reviewer could verify by ear.

Caveat: this is inter-ASR *agreement*, not ground truth; Whisper is itself weaker in Telugu, so the Telugu figure is a loose upper bound on true error.

5.3 How reliable are the emotion tags? — cross-model agreement

Emotion is the hardest dimension. I re-tagged **120 clips (60/language)** with a larger model, **sarvam-105b**, and compared to the shipped **sarvam-30b** labels:

Metric	Emotion	Style
Agreement	65 %	58 %
Cohen's κ	0.555 (moderate)	0.404

Per language, emotion agreement: English 63 %, Telugu 67 %. Moderate κ means the tags are *reasonably reproducible* but imperfect — an honest result. The confusion matrix shows disagreement concentrates between adjacent states (calm $\leftrightarrow$ neutral, excited $\leftrightarrow$ happy), not gross errors. This is exactly why every row ships with `emotion_confidence` + `tag_source`, and why the human-review tool (which records true human labels) is part of the workflow.

5.4 Phonetic and lexical coverage

	Indian English	Telugu
Distinct phonemes	39 / 39 (100 %)	45 / ~50 (90 %)
Unique words	1,553	2,072
Type–token ratio	0.33	0.59
Speaking rate (median WPM)	149	117

Full English phoneme coverage and ~90 % Telugu coverage from just 30 min each — the storytelling/audiobook source mix pays off. (g2p_en for English, epitran for Telugu; Telugu's higher TTR reflects its agglutinative morphology and the diverse novel/story text.)

5.5 Segmentation and gate evidence

- **Edge silence:** mean leading/trailing $\approx$ 0.0 s, internal silence **5.9 %** — tight, consistent cuts.
- **Quality-gate ablation:** median SNR (English) rose **30.5 $\rightarrow$ 31.6 dB** after gates (12 low-SNR clips removed); Telugu sources were already clean (0 rejects). Kept-clip median SNR $\approx$ 28–32 dB.

All claims above are reproducible: `scripts/eval_speaker.py`, `eval_asr.py`, `eval_emotion.py`, `eval_phoneme.py`, `eval_basic.py`; figures in the Appendix; raw numbers in `data/manifests/eval_*.json`.

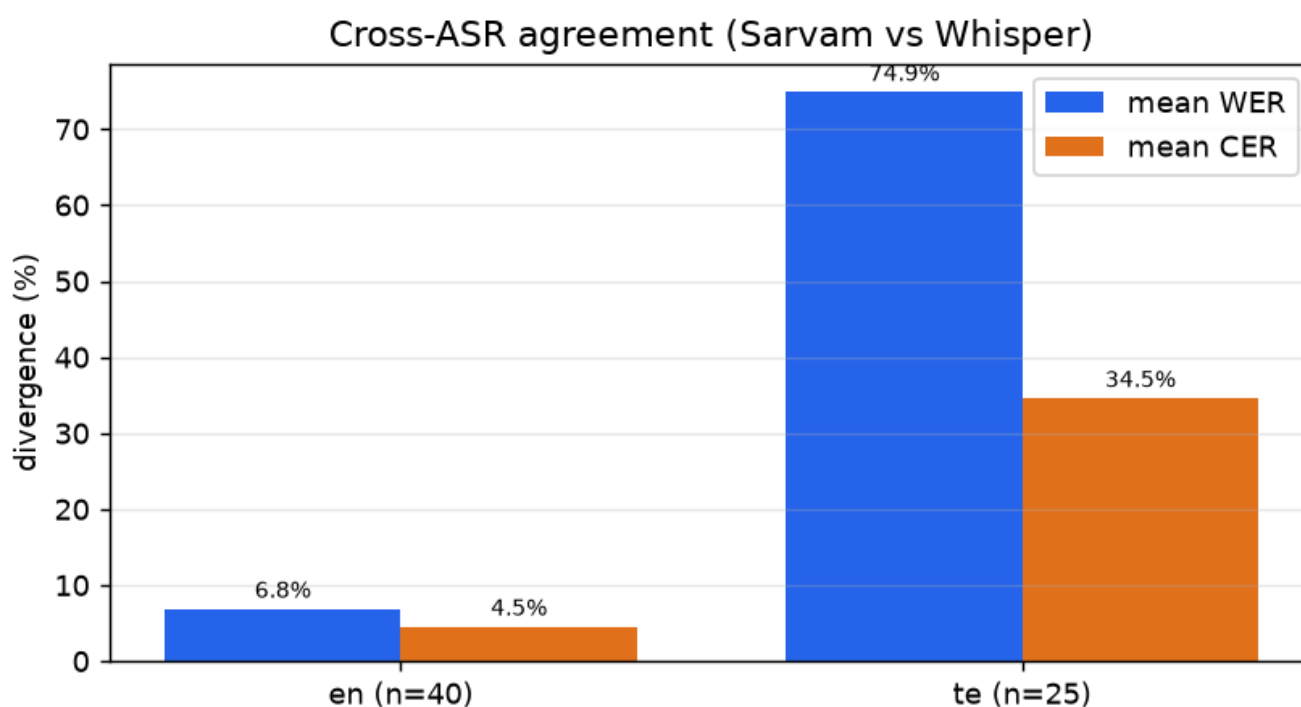
6. What I'd improve with more time

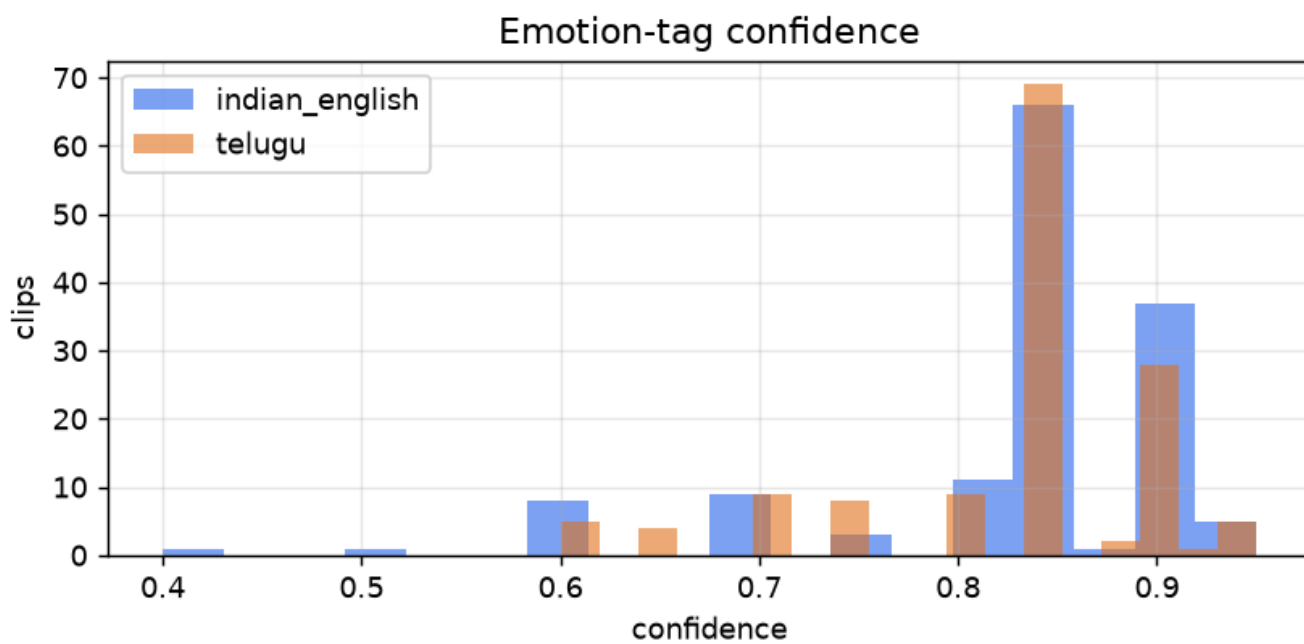
- **A true human evaluation pass** — the automated checks in §5 are proxies. With listening time I'd compute real human WER on a 100-clip audit and human inter-annotator agreement on emotion (the review tool already collects exactly this); that converts the proxy metrics into ground-truth numbers.
- **A trained SER model** (speech-emotion-recognition) as a third opinion alongside the acoustic+LLM tagger, with majority agreement used to auto-confirm and disagreement routed to review — this would lift emotion κ beyond the current 0.55.
- **Forced-alignment** (WhisperX/MFA) for word-level boundaries to trim even more tightly and to auto-flag transcript/audio mismatches.
- **Background-music separation** (Demucs) to rescue otherwise-good clips that carry a light bed.
- **Language-aware text normalization** (number/abbreviation expansion) for `normalized_text`.
- **A larger, more balanced source pool** per emotion (happy/surprised are the thinnest buckets).

Links - Dataset: <https://huggingface.co/datasets/AkCodes23/sarvam-tts-in-te-en> - Code: <https://github.com/AkCodes23/Sarvam-AI>

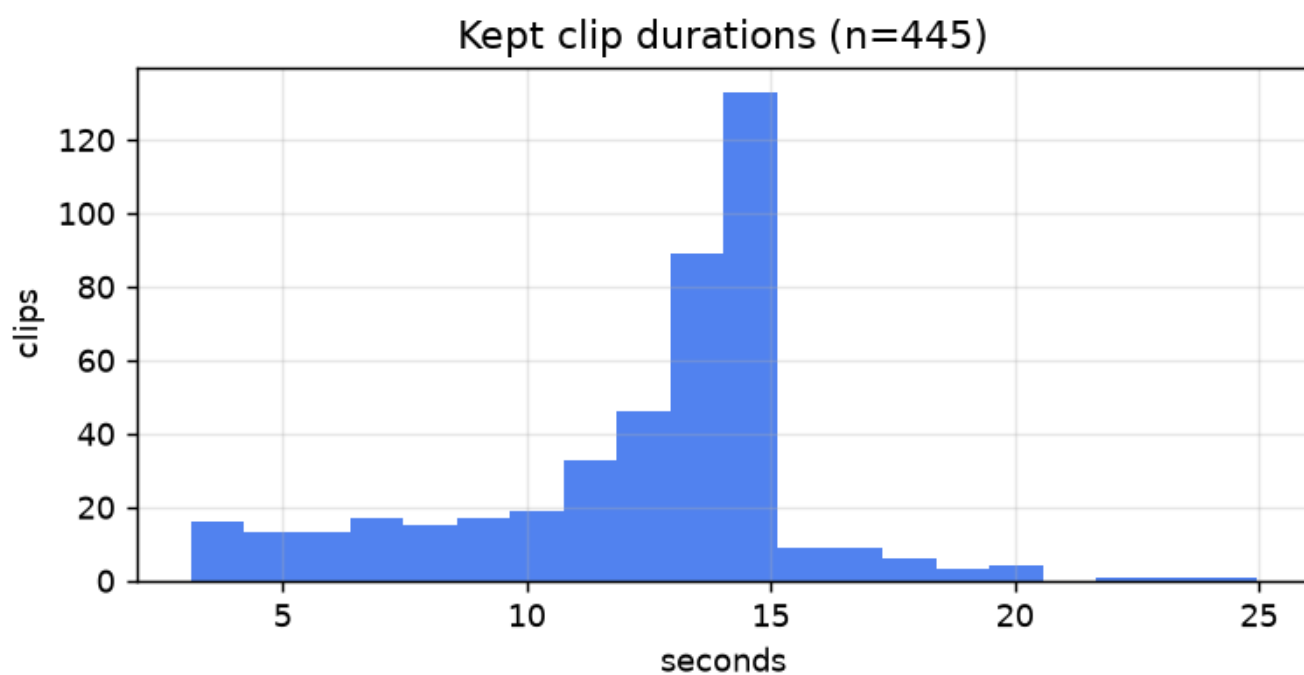
Appendix: figures are generated by `scripts/make_figures.py`; all thresholds live in `config/config.yaml`; the dataset card is in `reports/DATASET_CARD.md`.

Appendix: Figures

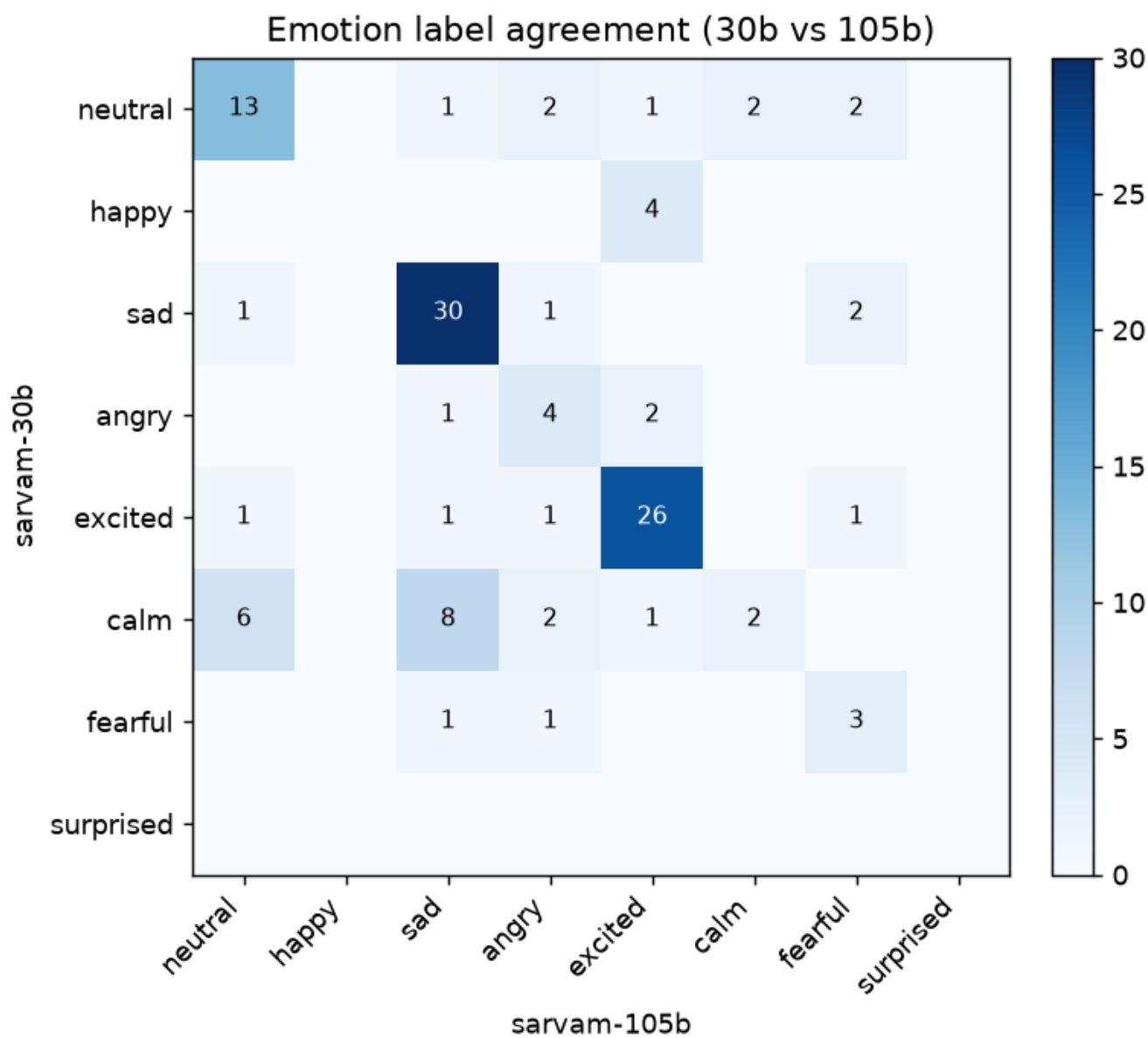




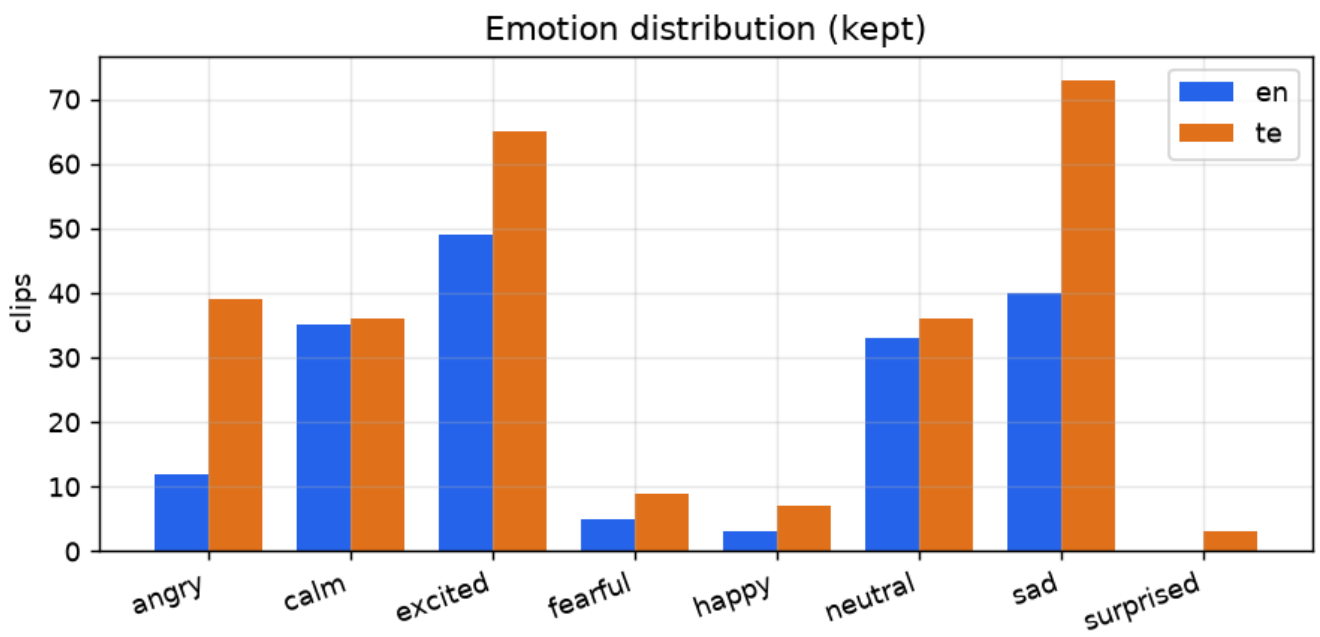
confidence_hist



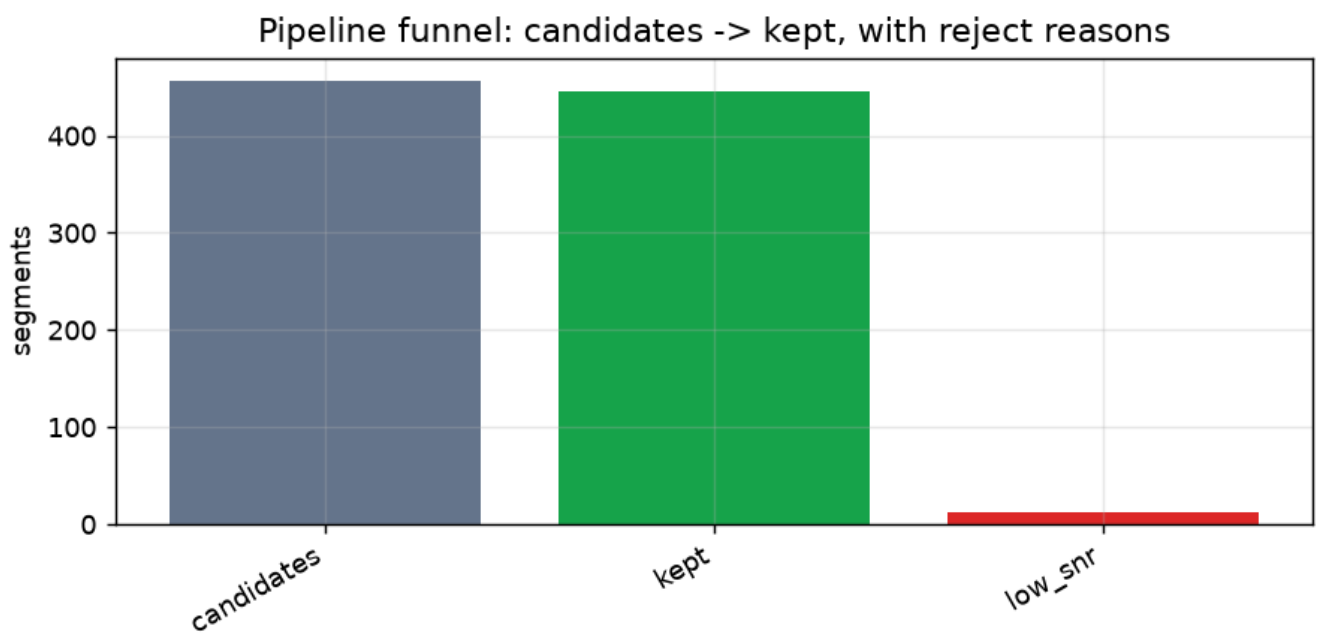
duration_hist



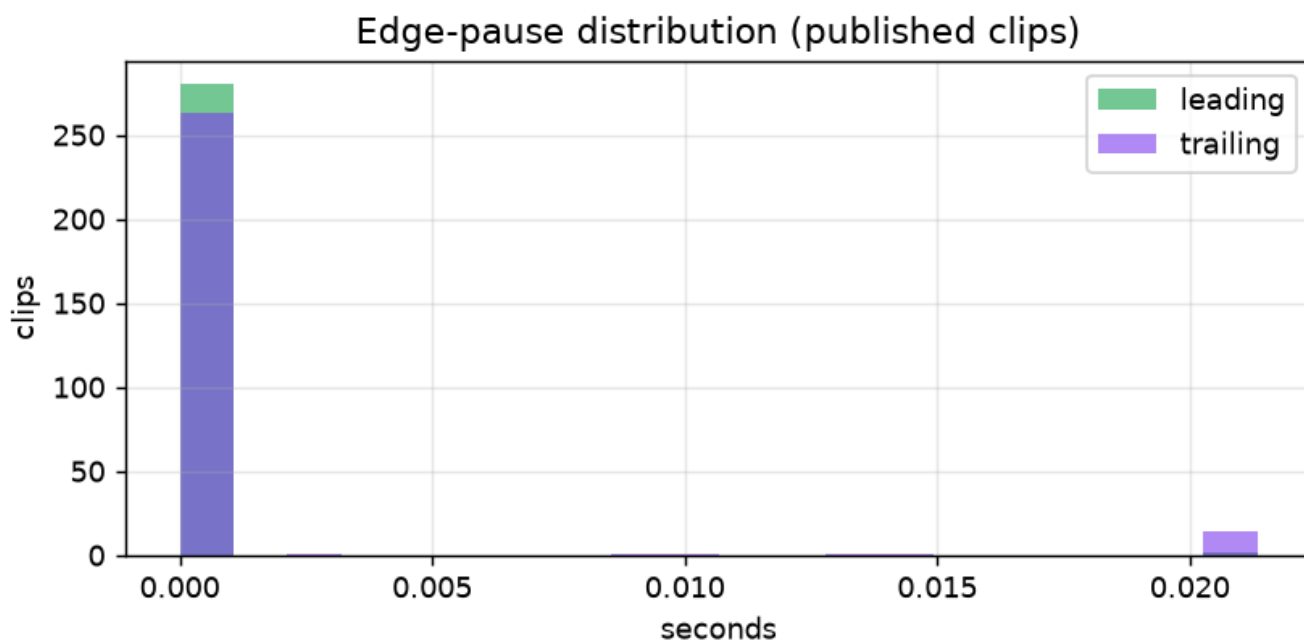
emotion_confusion



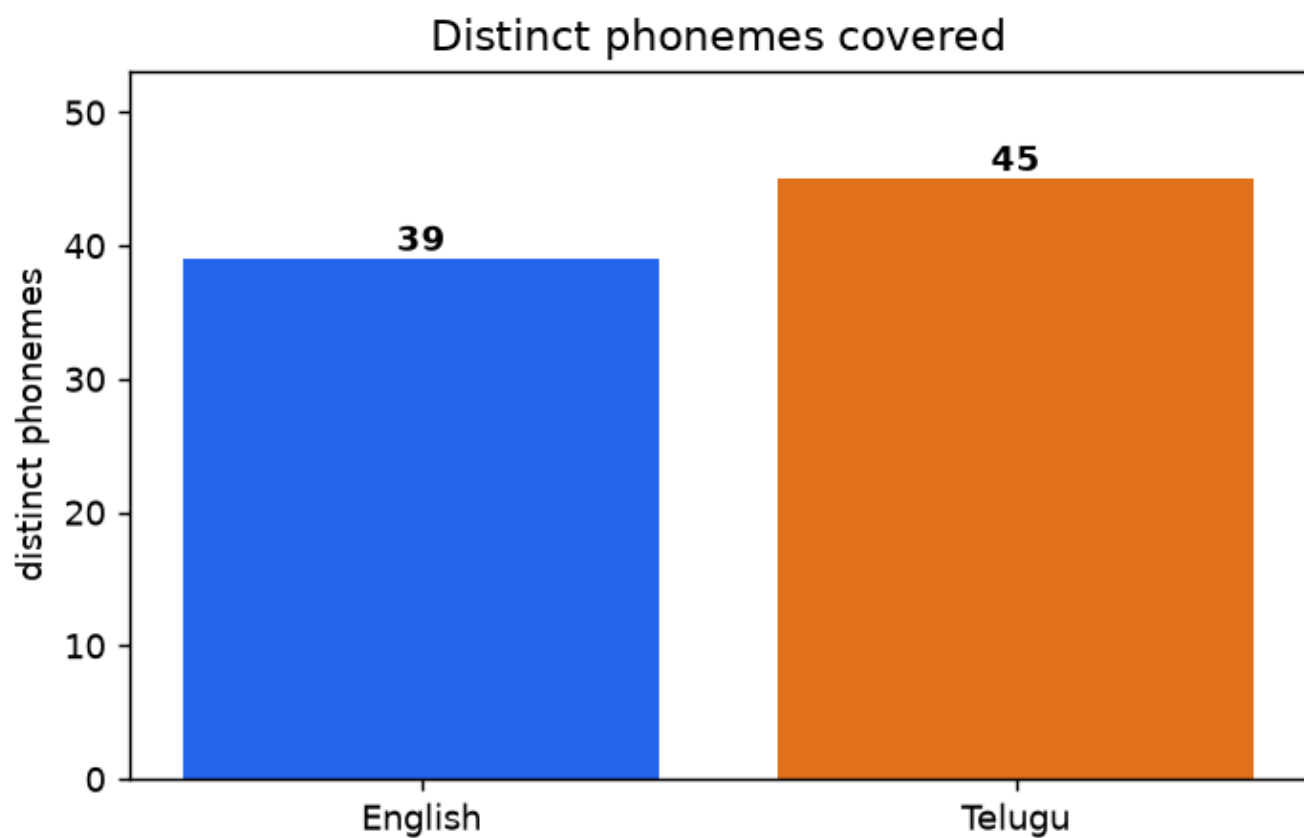
emotion_dist



funnel

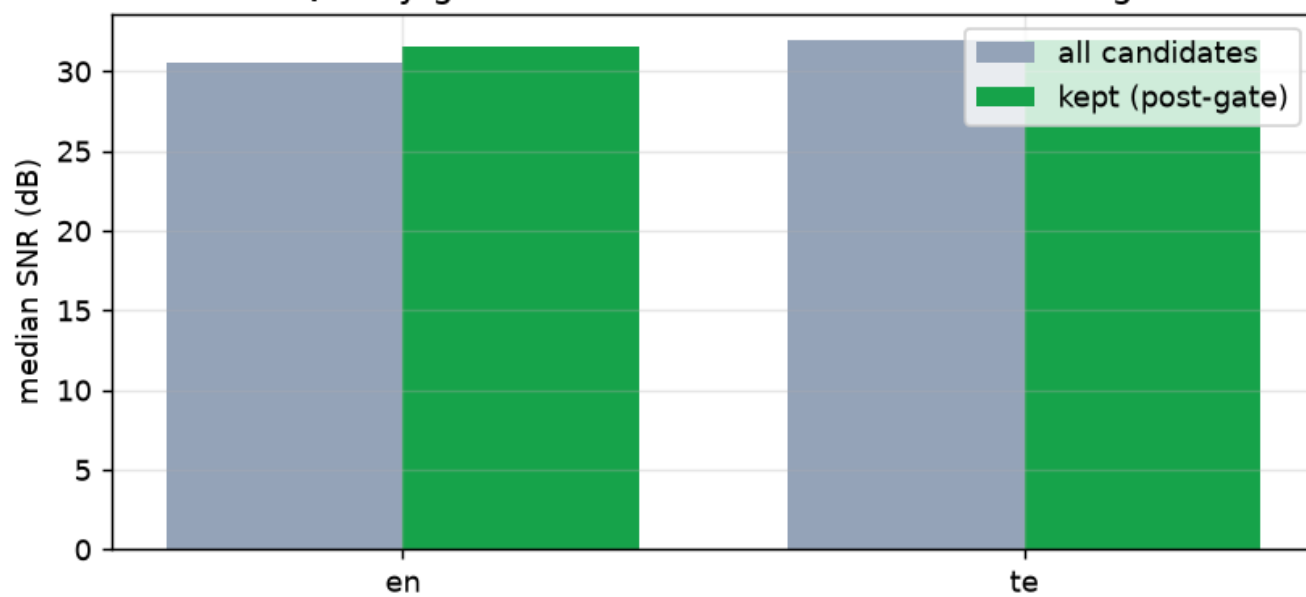


pause_dist



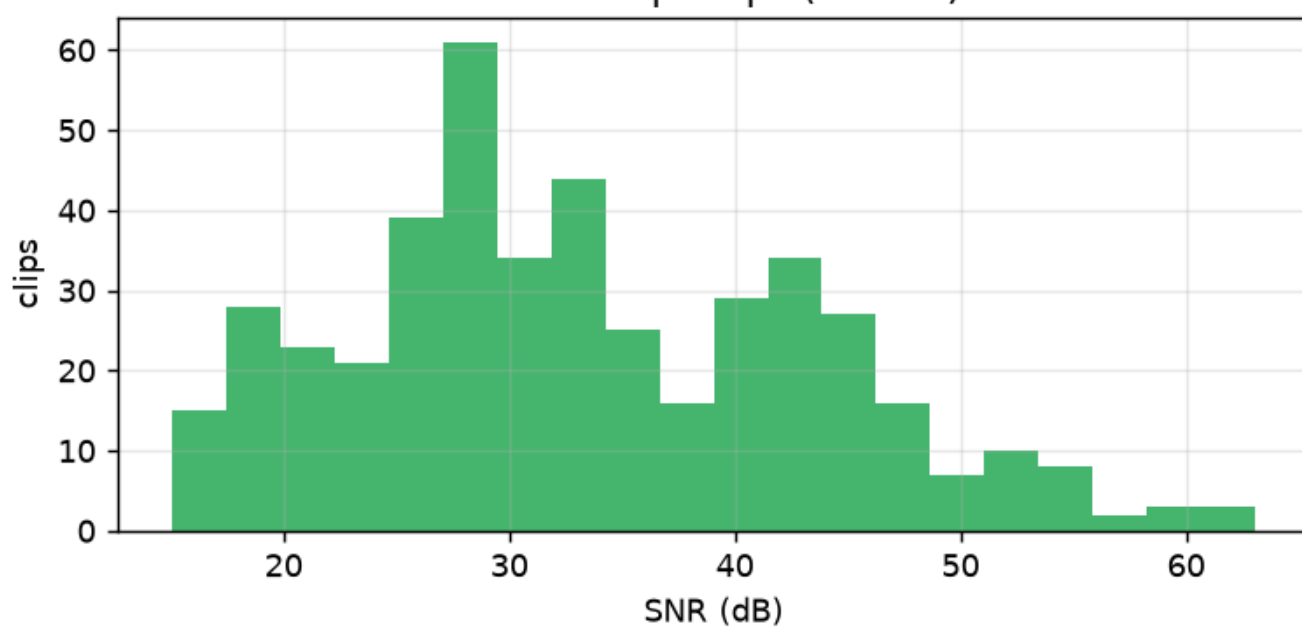
phoneme_coverage

Quality-gate ablation: SNR before/after filtering

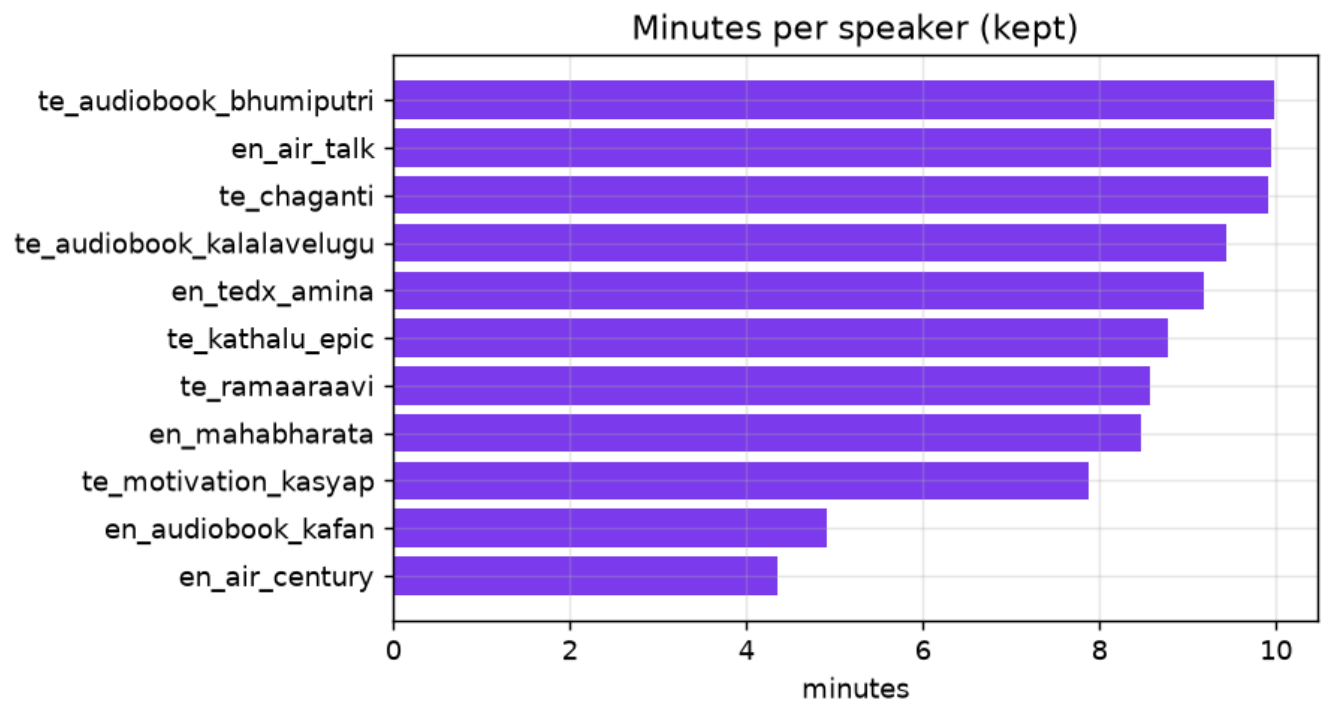


snr_ablation

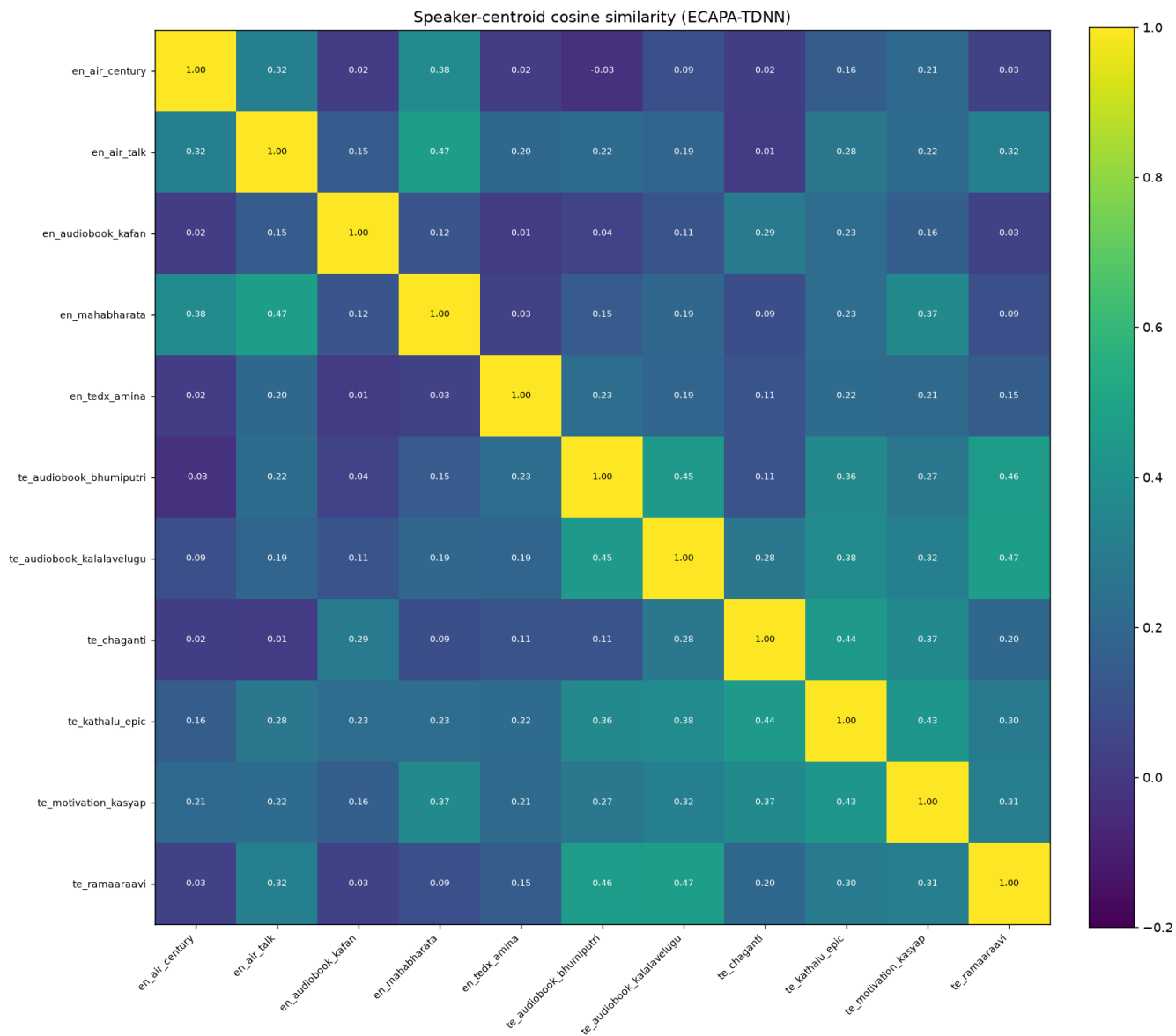
SNR of kept clips (n=445)



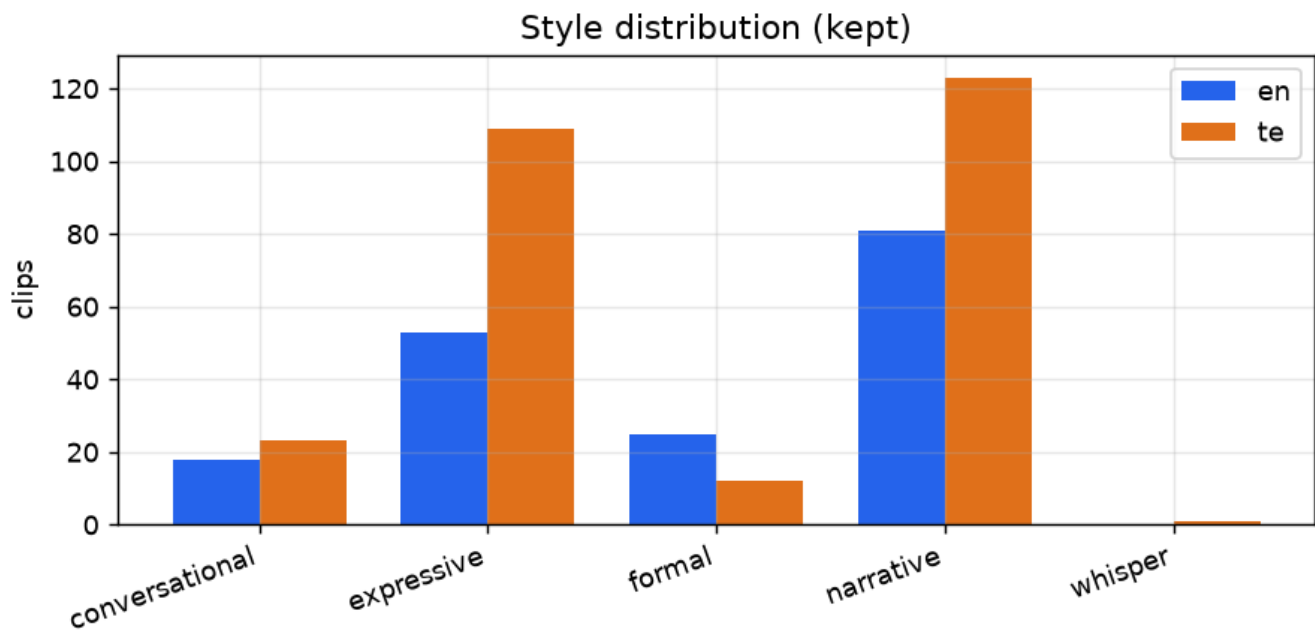
snr_hist



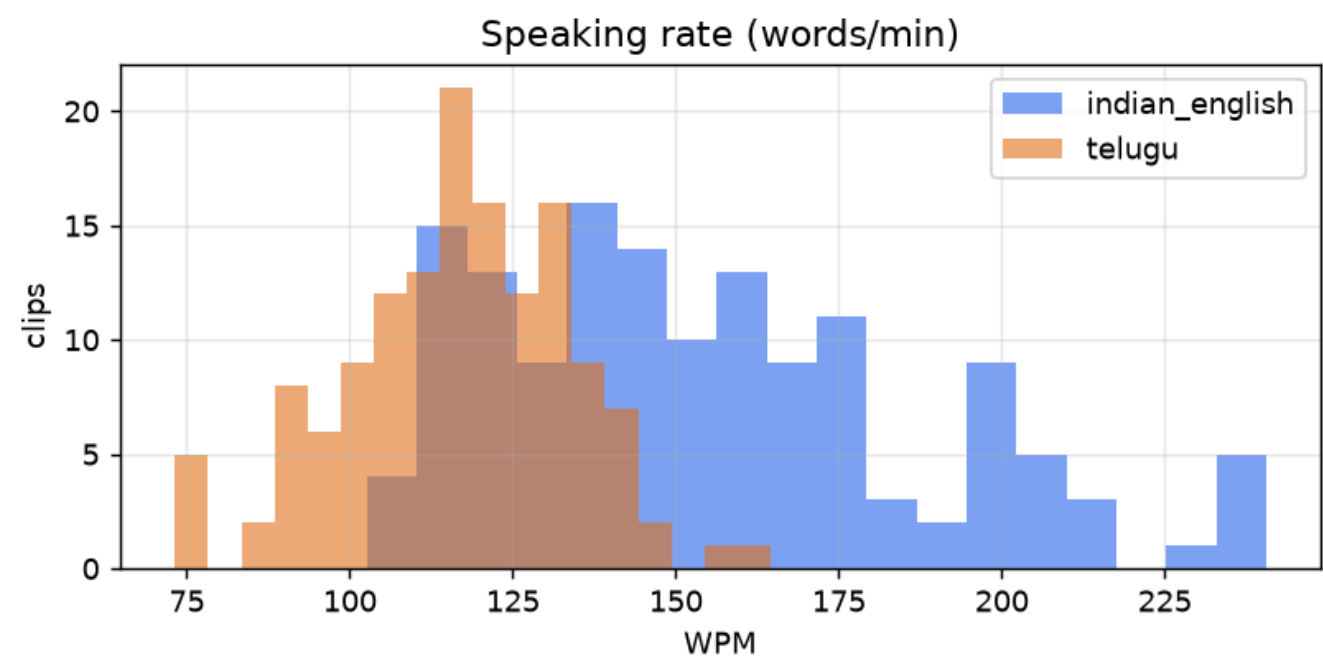
speaker_minutes



speaker_similarity



style_dist



wpm_hist